Area 4: Power Transmission

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

Contents

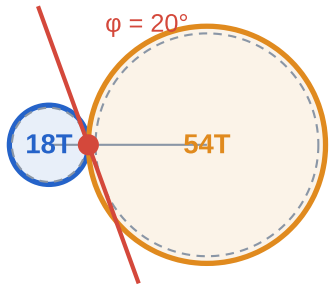
4.A Gears and Gear Trains
4.B Bearings
4.C Belts, Chains, Clutches, Brakes, Power Screws
4.D Shafts and Keys
4.E Motors and Engines

4.A Gears and Gear Trains

1. Key concepts

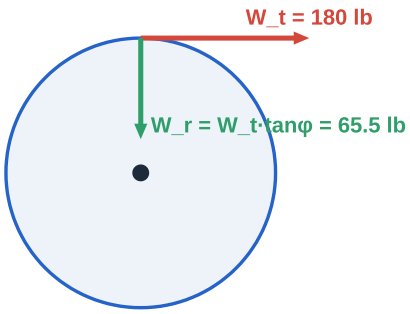
- Geometry: diametral pitch P = N/d (teeth/in) or module m = d/N (mm). Center distance = $(d_1 + d_2)/2$. Mating gears must share P (or m) and the pressure angle.
- Forces (spur): $W_t = 33{,}000hp/V$ (V in ft/min), radial $W_r = W_t \cdot \tan\varphi$. Helical: axial $W_a = W_t \cdot \tan\psi$.
- Train value = product of driver teeth / product of driven teeth. Planetary: use the Willis formula $(n_L - n_A)/(n_F - n_A) = e$.
- Bending (Lewis): $\sigma = K_v W_t P/(FY)$. Wear (contact) uses the AGMA equations.

Diagrams



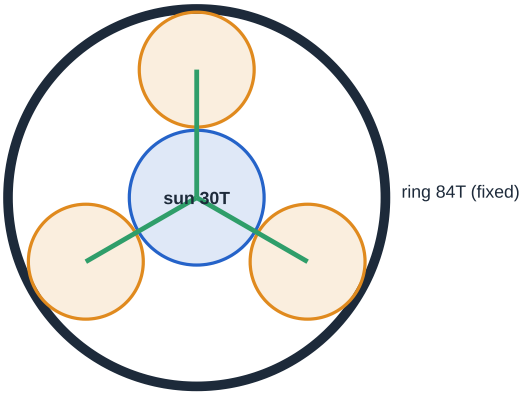
P = 6; d_1 = 3 in, d_2 = 9 in; C = 6 in

Spur gear mesh: pitch circles, pitch point, line of action.



helical: $W_a = W_t \cdot \tan\psi$ = 65.5 lb (ψ = 20°)

Tooth forces: W_t, $W_r = W_t \tan\varphi$, $W_a = W_t \tan\psi$.



carrier (green) = output; $n_{arm} = n_{sun} \cdot N_s/(N_s+N_r)$

Planetary train: sun, planets, ring, carrier.

2. Essential formulas

Name	Formula	Variables / units
Pitch	$P = N/d\ ;\ m = d/N$	
Pitch-line velocity	$V = \pi dn/12\ (\text{ft/min})$	
Tangential load	$W_t = 33{,}000\ hp/V$	
Radial	$W_r = W_t \tan\varphi$	
Helical axial	$W_a = W_t \tan\psi$	

Lewis	$\sigma = K_v W_t P/(FY)$	
Planetary	$(n_{ring} - n_{arm})/(n_{sun} - n_{arm}) = -N_{sun}/N_{ring}$	

3. Handbook search keywords

diametral pitch | gear forces | Lewis bending | gear train value | planetary gear train

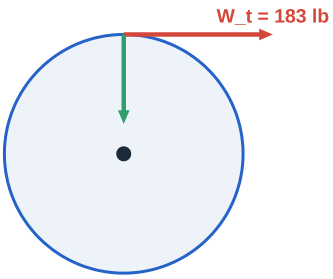
4. Worked examples

Example 1: Spur gear forces
An 18T pinion with P = 6 and φ = 20° transmits 5 hp at 1,150 rpm. Find W_t and W_r.
Handbook search: **gear tooth forces**

1. d = 18/6 = 3 in; V = π(3)(1,150)/12 = 903 ft/min.
2. W_t = 33,000(5)/903 = 182.7 lb; W_r = 66.5 lb.

Answer: $W_t \approx 183$ lb, $W_r \approx 66$ lb

Check your answer: $W_r = W_t \cdot \tan 20° \approx 0.364\, W_t$.

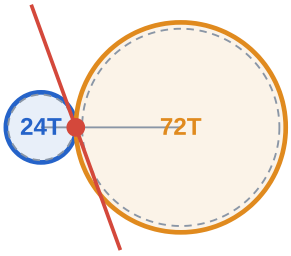


Try it yourself: Center distance
A 24T pinion meshes with a 72T gear, P = 4 teeth/in. Find the center distance and the speed ratio.
Handbook search: **center distance**

1. d_1 = 24/4 = 6 in; d_2 = 72/4 = 18 in.
2. C = $(d_1 + d_2)/2$ = 12 in; ratio = 72/24 = 3.

Answer: C = 12 in; ratio 3:1

Check your answer: C = $(N_1 + N_2)/(2P)$ gives the same result in one step.



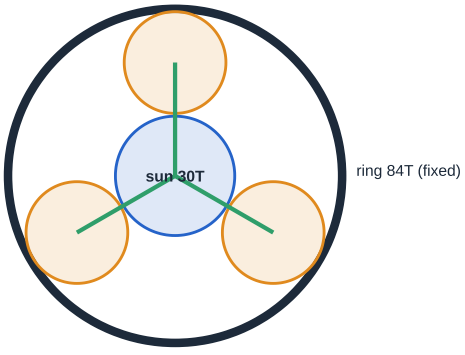
P = 4; d_1 = 6 in, d_2 = 18 in; C = 12 in

Example 3: Planetary carrier speed
Sun 30T input at 1,200 rpm, ring 84T fixed, carrier output. Find the carrier speed.
Handbook search: **planetary gear train**

1. With the ring fixed: $n_c = n_s \cdot N_s/(N_s + N_r)$ = 1,200 × 30/114 = 315.8 rpm.

Answer: 315.8 rpm

Check your answer: The carrier turns the same way as the sun but slower. The ratio $1 + N_r/N_s$ = 3.8.



carrier (green) = output; $n_{arm} = n_{sun} \cdot N_s/(N_s+N_r)$

5. Common traps

- Using V in ft/s.
- Mixing up driver and driven teeth in the train value.
- Using the pressure angle for the helical thrust (use the helix angle).

4.A Gears and Gear Trains: one-page cheat summary

- ► $P = N/d$; $C = (N_1 + N_2)/(2P)$
- ► $V = \pi dn/12$
- ► $W_t = 33{,}000hp/V$
- ► $W_r = W_t \tan\varphi$; $W_a = W_t \tan\psi$
- ► Planetary: Willis formula

Formulas

Pitch	$P = N/d$; $m = d/N$
Pitch-line velocity	$V = \pi dn/12$ (ft/min)
Tangential load	$W_t = 33{,}000\ hp/V$
Radial	$W_r = W_t \tan\varphi$
Helical axial	$W_a = W_t \tan\psi$
Lewis	$\sigma = K_v W_t P/(FY)$
Planetary	$(n_{ring} - n_{arm})/(n_{sun} - n_{arm}) = -N_{sun}/N_{ring}$

Search keywords

diametral pitch, gear forces, Lewis bending, gear train value, planetary gear train

Top traps

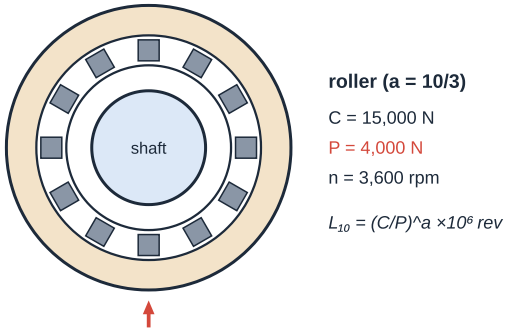
- Using V in ft/s.
- Mixing up driver and driven teeth in the train value.
- Using the pressure angle for the helical thrust (use the helix angle).

4.B Bearings

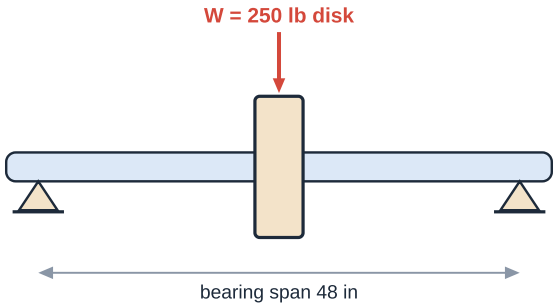
1. Key concepts

- Rolling bearings: $L_{10} = (C/P)^a \times 10^6$ revolutions, with a = 3 for ball and 10/3 for roller bearings. Hours = rev/(60n).
- Equivalent load P = XF_r + YF_a when F_a/F_r > e. Required rating C = P(L_D/L_R)^(1/a).
- Reliability adjustment (Weibull): for R ≠ 0.90, the life scales with [ln(1/R)/ln(1/0.9)]^(1/b).
- Journal bearings: Sommerfeld number S = (r/c)²μN/P; the Petroff torque estimates friction.

Diagrams



Rolling bearing: C rating vs equivalent load P.



Shaft on two bearings carrying a central load.

2. Essential formulas

Name	Formula	Variables / units
Rating life	$L_{10} = (C/P)^a \times 10^6$ rev	
Hours	$L_h = L_{10}/(60n)$	
Equivalent load	$P = XF_r + YF_a$	
Required C	$C = P(L_h 60n/10^6)^{(1/a)}$	
Sommerfeld	$S = (r/c)^2 \mu N/P$	

3. Handbook search keywords

bearing L10 life · dynamic load rating · equivalent radial load · Sommerfeld number

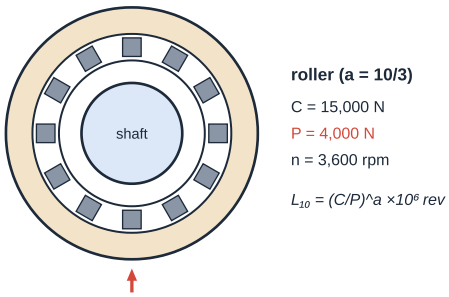
4. Worked examples

Example 1: Roller bearing life
Roller bearing with C = 15,000 N, P = 4,000 N, n = 3,600 rpm. Find L_{10} in hours.
Handbook search: **L10 life**

1. $L_{10} = (15{,}000/4{,}000)^{(10/3)} \times 10^6$ rev = 81,928,982 rev.
2. Hours = rev/(60 × 3,600) = 379 h.

Answer: 379 h

Check your answer: Roller exponent 10/3, ball exponent 3. Doubling P cuts life by about 10×.



Try it yourself: Required rating C

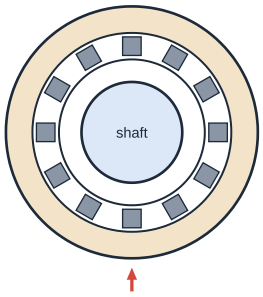
A roller bearing needs L_{10} = 30,000 h at 1,200 rpm under P = 8,000 N. Find C.

Handbook search: **basic dynamic load rating**

1. L (rev) = 30,000 × 60 × 1,200 = 2,160 × 10^6 rev.
2. C = P·L^(3/10) = 80,062 N.

Answer: 80,062 N

Check your answer: Put C back in: (C/P)^(10/3) × 10^6/(60n) should return 30,000 h.



roller (a = 10/3)

C = 80,062 N

P = 8,000 N

n = 1,200 rpm

$L_{10} = (C/P)^a \times 10^6$ rev

Example 3: Combined radial and thrust

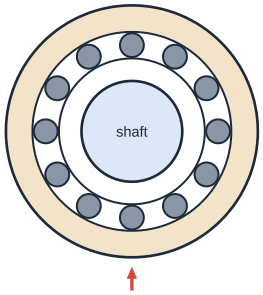
A ball bearing (C = 29,000 N) has F_r = 5,000 N and F_a = 3,000 N at 1,800 rpm; X = 0.56, Y = 1.6. Find L_{10} (h).

Handbook search: **equivalent load**

1. P = XF_r + YF_a = 7,600 N.
2. $L_{10} = (C/P)^3 \times 10^6/(60n)$ = 514 h.

Answer: 514 h

Check your answer: P must be at least F_r. If it comes out below 5,000 N, re-check X and Y.



ball (a = 3)

C = 29,000 N

P = 7,600 N

n = 1,800 rpm

$L_{10} = (C/P)^a \times 10^6$ rev

5. Common traps

- Using a = 3 for roller bearings.
- Forgetting the 60 min/h.
- Using the radial load alone when thrust is present.

4.B Bearings: one-page cheat summary

- ► $L_{10} = (C/P)^a\ 10^6$ rev
- ► a = 3 ball, 10/3 roller
- ► $P = XF_r + YF_a$
- ► $h = rev/60n$

Formulas

Rating life	$L_{10} = (C/P)^a \times 10^6$ rev
Hours	$L_h = L_{10}/(60n)$
Equivalent load	$P = XF_r + YF_a$
Required C	$C = P(L_h 60n/10^6)^{(1/a)}$
Sommerfeld	$S = (r/c)^2 \mu N/P$

Search keywords

bearing L10 life, dynamic load rating, equivalent radial load, Sommerfeld number

Top traps

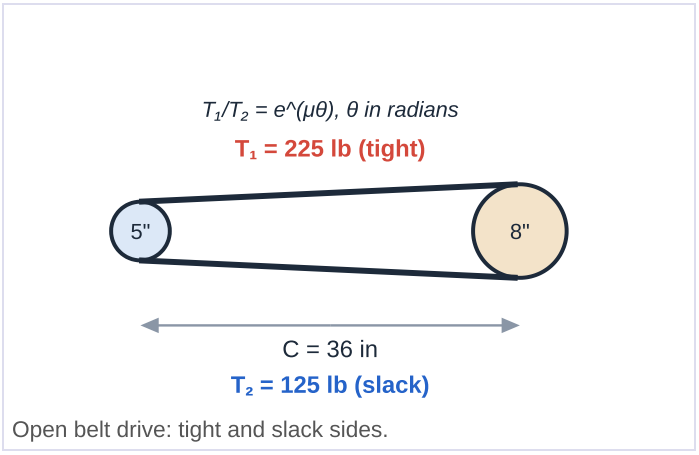
- Using a = 3 for roller bearings.
- Forgetting the 60 min/h.
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4.C Belts, Chains, Clutches, Brakes, Power Screws

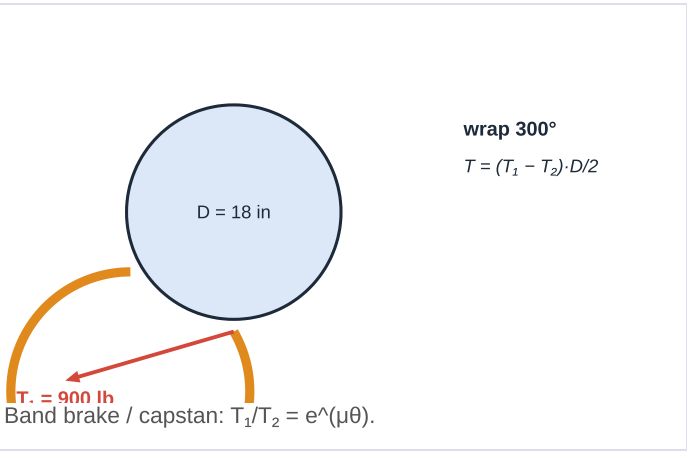
1. Key concepts

- Flat belt: $T_1/T_2 = e^{(\mu\theta)}$ at impending slip (V-belts use $\mu/\sin(\beta/2)$). Power = $(T_1 - T_2)V$.
- Chain: V = NpN_rpm/12 (ft/min). Chordal action is reduced by using more sprocket teeth.
- Disk clutch (uniform wear): T = NμF(D + d)/4. Uniform pressure gives slightly more torque. Band brake: T = $(T_1 - T_2)r$ with $T_1/T_2 = e^{(\mu\theta)}$.
- Square-thread power screw: T_raise = (Wd_m/2)(l + πμd_m)/(πd_m − μl) + collar torque Wμ_cd_c/2. It is self-locking if πμd_m > l. Efficiency = Wl/(2πT).

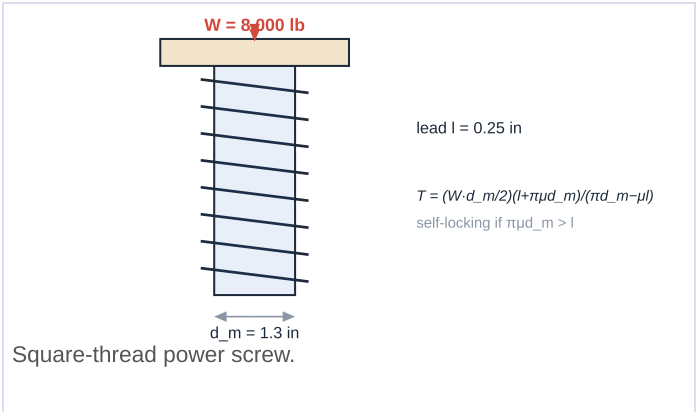
Diagrams



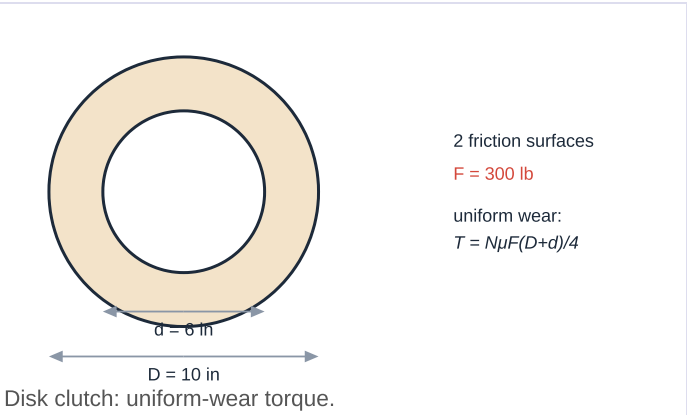
Open belt drive: tight and slack sides.



Band brake / capstan: $T_1/T_2 = e^{(\mu\theta)}$.



Square-thread power screw.



Disk clutch: uniform-wear torque.

2. Essential formulas

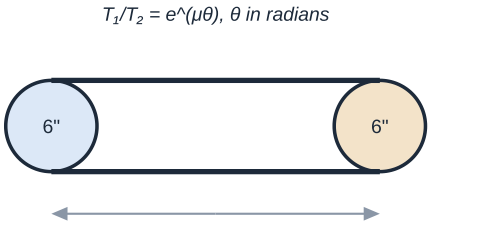
Name	Formula	Variables / units
Belt	$T_1/T_2 = e^{(\mu\theta)}$; $hp = (T_1 - T_2)V/33{,}000$	
Open-belt length	$L = 2C + \pi(D + d)/2 + (D - d)^2/(4C)$	
Clutch (uniform wear)	$T = N\mu F(D + d)/4$	
Screw raise	$T = (Wd_m/2)(l + \pi\mu d_m)/(\pi d_m - \mu l)$	
Efficiency	$e = Wl/(2\pi T)$	

3. Handbook search keywords

belt tension ratio · band brake · disk clutch uniform wear · power screw torque · chain velocity

4. Worked examples

Example 1: Flat belt tensions
A belt transmits 5 hp from a 6 in pulley at 1,150 rpm, θ = 180°, μ = 0.3. Find T_1.
Handbook search: **belt friction**

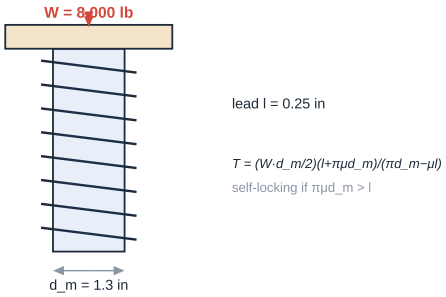


1. Net force $T_1 - T_2 = T/r = (63{,}025 \cdot 5/1{,}150)/3 = 91.3$ lb.
2. $e^{(\mu\theta)} = e^{(0.3\pi)} = 2.566$; $T_1 = F \cdot e/(e - 1) = 149.7$ lb.

Answer: 150 lb

Check your answer: $T_1 - T_2$ must equal 2T/D, and $T_1/T_2 = e^{(\mu\theta)}$.

Try it yourself: Power screw raise torque
Square thread, d_m = 1.3 in, lead 0.25 in, μ = 0.12, raising 8,000 lb (no collar). Find T and efficiency.
Handbook search: **power screw**

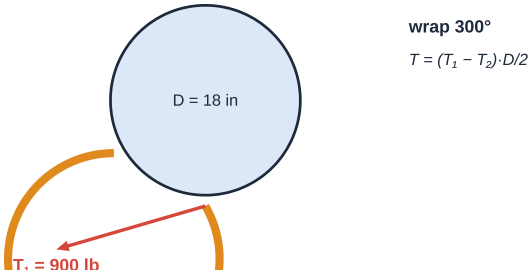


1. $T_R = (W\, d_m/2)(l + \pi\mu d_m)/(\pi d_m - \mu l) = 949$ in-lb.
2. $e = Wl/(2\pi T) = 33.5\%$.
3. Self-locking since $\mu = 0.12 > \tan\lambda = 0.061$.

Answer: T ≈ 949 in-lb; e ≈ 34%

Check your answer: A self-locking square thread always has efficiency below 50%.

Example 3: Band brake torque
18 in drum, 300° wrap, μ = 0.3, max band tension 900 lb. Find the braking torque.
Handbook search: **band brake**



1. $T_2 = T_1 e^{(-\mu\theta)} = 900 \times e^{(-0.3 \times 5.236)} = 187$ lb.
2. Torque $= (T_1 - T_2)r = (900 - 187) \times 9 = 6{,}416$ in-lb.

Answer: 6,416 in-lb

Check your answer: θ must be in radians inside the exponent.

5. Common traps

- Using θ in degrees in $e^{(\mu\theta)}$.
- Forgetting the collar friction on a power screw.
- Using uniform pressure when the question says uniform wear.

4.C Belts, Chains, Clutches, Brakes, Power Screws: one-page cheat summary

- ► $T_1/T_2 = e^{(\mu\theta)}$, θ in rad
- ► $hp = (T_1 - T_2)V/33{,}000$
- ► Clutch: $N\mu F(D + d)/4$
- ► Screw: self-locking if $\pi\mu d_m > l$

Formulas

Belt	$T_1/T_2 = e^{(\mu\theta)}$; $hp = (T_1 - T_2)V/33{,}000$
Open-belt length	$L = 2C + \pi(D + d)/2 + (D - d)^2/(4C)$
Clutch (uniform wear)	$T = N\mu F(D + d)/4$
Screw raise	$T = (Wd_m/2)(l + \pi\mu d_m)/(\pi d_m - \mu l)$
Efficiency	$e = Wl/(2\pi T)$

Search keywords

belt tension ratio, band brake, disk clutch uniform wear, power screw torque, chain velocity

Top traps

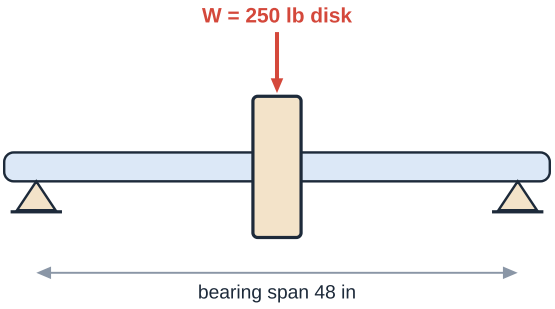
- Using θ in degrees in e^(μθ).
- Forgetting the collar friction on a power screw.
- Using uniform pressure when the question says uniform wear.

4.D Shafts and Keys

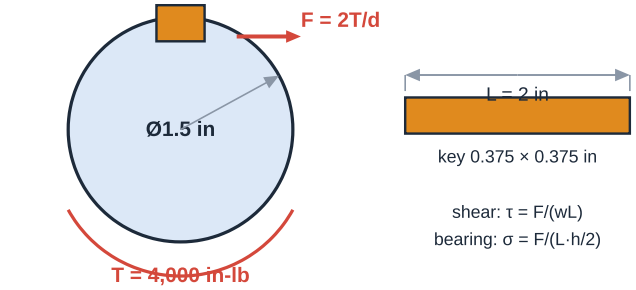
1. Key concepts

- Design steps: loads → shear and moment diagrams → critical sections (shoulders, keyways) → fatigue size → check deflection, slopes and critical speed.
- DE-Goodman with rotating bending (M_a) and steady torque (T_m): $d = \{16n/\pi\,[2K_{fM}M_a/S_e + \sqrt{3}K_{fs}T_m/S_{ut}]\}^{1/3}$.
- Critical speed: $\omega = \sqrt{g/\delta_{st}}$ for a single disk (Rayleigh for several). Keep the operating speed well away from it.
- Flywheel: $\Delta E = C_s I\omega^2$, with $C_s = (\omega_{max} - \omega_{min})/\omega_{avg}$.

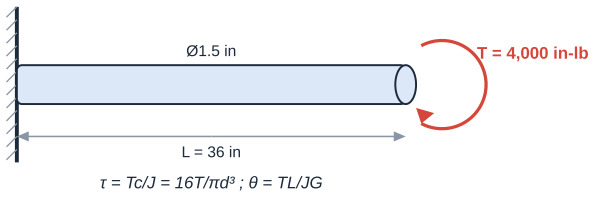
Diagrams



Disk at midspan: $\delta = WL^3/48EI$ sets the first critical speed.



Keyed shaft section.



Torsional load on the shaft.

2. Essential formulas

Name	Formula	Variables / units
DE-Goodman diameter	$d = \{16n/\pi\,[2K_{fM}M_a/S_e + \sqrt{3}K_{fs}T_m/S_{ut}]\}^{1/3}$	
Static torsion	$d = (16Tn/(\pi S_{sy}))^{1/3}$	
Critical speed	$n_c = (30/\pi)\sqrt{(g/\delta)}$	$g = 386$ in/s²
Flywheel	$\Delta E = C_s I\omega^2$	

3. Handbook search keywords

shaft design DE-Goodman · critical speed · flywheel coefficient of fluctuation · key

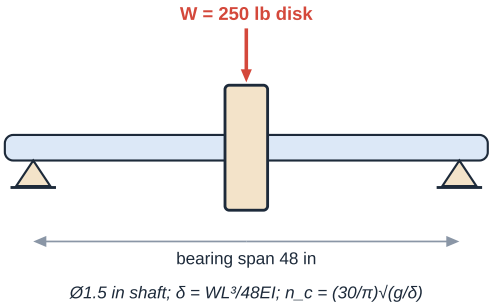
4. Worked examples

Example 1: Critical speed
A 250 lb disk sits at the center of a Ø1.5 in steel shaft (E = 30×10⁶ psi) on bearings 48 in apart. Find n_c.
Handbook search: **critical speed**

1. I = π(1.5)⁴/64 = 0.2485 in⁴; δ = WL³/48EI = 0.0773 in.
2. n_c = (30/π)√(386.1/δ) = 675 rpm.

Answer: 675 rpm

Check your answer: Run the shaft well below n_c (or well above with care). A rule of thumb is a 20% margin.



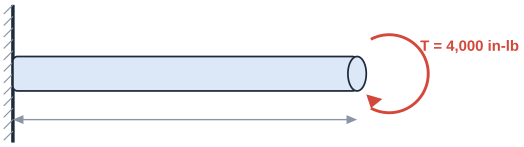
Ø1.5 in shaft; δ = WL³/48EI; n_c = (30/π)√(g/δ)

Try it yourself: DE-Goodman shaft size
M_a = 3,000 in-lb (reversed), T_m = 4,000 in-lb, K_f = 1.8, K_fs = 1.3, Se = 30 ksi, Sut = 80 ksi, n = 1.5. Find d.
Handbook search: **DE-Goodman**

1. d = {16n/π [2K_fM_a/S_e + √3·K_fsT_m/S_ut]}^(1/3).
2. d = 1.534 in.

Answer: 1.534 in

Check your answer: Round up to the next standard size. Re-check K_f at the new diameter.



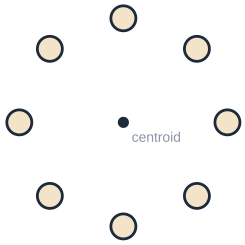
τ = Tc/J = 16T/πd³ ; θ = TL/JG

Example 3: Flange coupling bolts
28,000 in-lb through 8 bolts, Ø0.5 in, on a 5 in bolt circle. Find the bolt shear stress.
Handbook search: **flange coupling**

1. Force per bolt = T/(n·r) = 28,000/(8 × 2.5) = 1,400 lb.
2. τ = F/A = 1,400/0.1963 = 7,130 psi.

Answer: 7,130 psi

Check your answer: Use the bolt-circle RADIUS (2.5 in), not the diameter.



F = P/n (direct) + M·r/Σr² (moment), added as vectors

5. Common traps

- Using g = 32.2 ft/s² with δ in inches.
- Leaving out K_f at shoulders.
- Applying the mean-torsion term to the alternating stress.

4.D Shafts and Keys: one-page cheat summary

- ► DE-Goodman d formula
- ► $n_c = (30/\pi)\sqrt{(g/\delta)}$
- ► $\Delta E = C_s I \omega^2$
- ► Check slopes at bearings

Formulas

DE-Goodman diameter	$d = \{16n/\pi\ [2K_f M_a/S_e + \sqrt{3}K_{fs}T_m/S_{ut}]\}^{(1/3)}$
Static torsion	$d = (16Tn/(\pi S_{sy}))^{(1/3)}$
Critical speed	$n_c = (30/\pi)\sqrt{(g/\delta)}$
Flywheel	$\Delta E = C_s I \omega^2$

Search keywords

shaft design DE-Goodman, critical speed, flywheel coefficient of fluctuation, key

Top traps

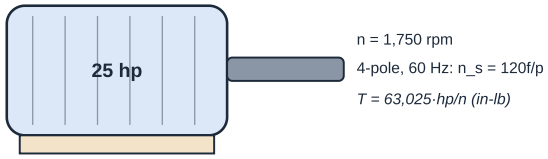
- Using g = 32.2 ft/s² with δ in inches.
- Leaving out K_f at shoulders.
- Applying the mean-torsion term to the alternating stress.

4.E Motors and Engines

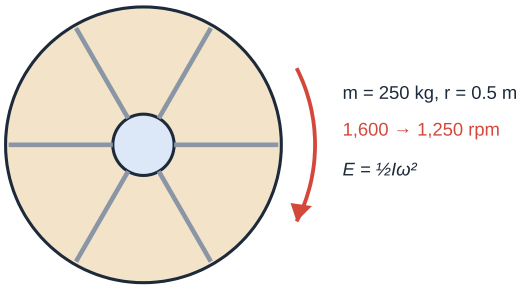
1. Key concepts

- Induction motor: synchronous speed n_s = 120f/p (p = poles). Slip s = (n_s − n)/n_s.
- Three-phase input: P = √3·V·I·pf. Output = η × input. T = 63,025hp/n.
- Engines: brake power = 2πNT. Thermal efficiency = output/fuel energy. BSFC = fuel rate/brake power.

Diagrams



Motor: power, torque, speed, slip.



Flywheel energy: $\Delta E = \tfrac{1}{2} I(\omega_1^2 - \omega_2^2)$.

2. Essential formulas

Name	Formula	Variables / units
Sync speed	$n_s = 120f/p$	
Slip	$n = n_s(1 - s)$	
3-phase power	$P = \sqrt{3} \cdot V \cdot I \cdot pf$	
Line current	$I = hp \cdot 746/(\sqrt{3} \cdot V \cdot \eta \cdot pf)$	

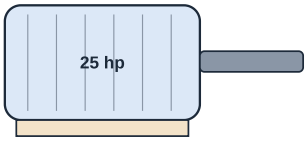
3. Handbook search keywords

induction motor slip · synchronous speed · three phase power · motor efficiency

4. Worked examples

Example 1: Motor current
A 25 hp, 460 V three-phase motor has η = 0.90 and pf = 0.85. Find the line current.

Handbook search: **three phase power**

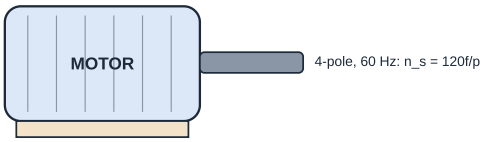


1. I = 25(746)/(√3·460·0.90·0.85) = 30.6 A.

Answer: 30.6 A

Check your answer: A rough rule is about 1.2–1.4 A per hp at 460 V.

Try it yourself: Slip and speed
A 4-pole, 60 Hz induction motor runs at 2% slip. Find the shaft speed.
Handbook search: **synchronous speed**

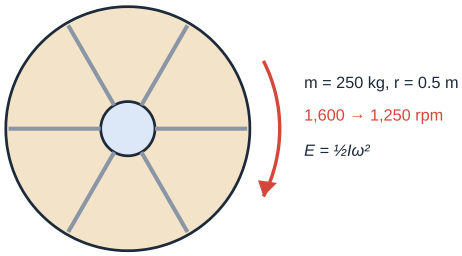


1. $n_s = 120f/p = 120 \times 60/4 = 1800$ rpm.
2. $n = n_s(1 - s) = 1764$ rpm.

Answer: 1764 rpm

Check your answer: Shaft speed must be slightly below synchronous speed.

Example 3: Flywheel energy release
A solid-disk flywheel (250 kg, r = 0.5 m) slows from 1,600 to 1,250 rpm. Find the energy released.
Handbook search: **flywheel**



1. $I = \frac{1}{2}mr^2 = 31.25$ kg·m².
2. $\Delta E = \frac{1}{2}I(\omega_1^2 - \omega_2^2) = 170.9$ kJ.

Answer: 170.9 kJ

Check your answer: Use rad/s, not rpm, inside ω².

5. Common traps

- Using poles/2 incorrectly ($n_s = 120f/p$ with p = total poles).
- Forgetting efficiency when going from output to input power.

4.E Motors and Engines: one-page cheat summary

- ► $n_s = 120f/p$
- ► $n = n_s(1 - s)$
- ► $P = \sqrt{3}VI \cdot pf$
- ► $T = 63{,}025hp/n$

Formulas

Sync speed	$n_s = 120f/p$
Slip	$n = n_s(1 - s)$
3-phase power	$P = \sqrt{3} \cdot V \cdot I \cdot pf$
Line current	$I = hp \cdot 746/(\sqrt{3} \cdot V \cdot \eta \cdot pf)$

Search keywords

induction motor slip, synchronous speed, three phase power, motor efficiency

Top traps

- Using poles/2 incorrectly ($n_s = 120f/p$ with p = total poles).
- Forgetting efficiency when going from output to input power.